Appendix A

The Gain to Pain Ratio

Most people tend to focus only on return. As I see it, as a performance measure, return is only meaningful relative to how much risk was required to get it. You could always get a higher return simply by using leverage; that doesn't mean it represents better performance. One statistic I particularly like is what I call the Gain to Pain ratio. I define the Gain to Pain ratio (GPR) as the sum of all monthly returns divided by the absolute value of the sum of all monthly losses.[1] This performance measure indicates the ratio of cumulative net gain to the cumulative loss realized to achieve that gain. For example, a GPR of 1.0 would imply that, on average, an investor would experience an equal amount of monthly losses to the *net* amount gained. If the average return per year is 12 percent (arithmetic, not compounded), the average amount of monthly losses per year would also sum to 12 percent. The GPR penalizes all losses in proportion to their size. Upside volatility, however, is beneficial because it only impacts the return portion of the ratio. In contrast, the Sharpe ratio—the most widely used return/risk measure—penalizes upside volatility. As a rough guideline, for liquid strategies, any GPR above 1.0 is very good, and a GPR above 1.5 is excellent.

[1]The Gain to Pain Ratio (GPR) is a performance statistic I have been using for many years. I am not aware of any prior use of this statistic, although the term is sometimes used as a generic reference for return/risk measures or a return/drawdown measure. The GPR is similar to the "profit factor," which is a commonly used statistic in evaluating trading systems. The profit factor is defined as the sum of all profitable trades divided by the absolute value of the sum of all losing trades. The profit factor is applied to trades, whereas the GPR is applied to interval (e.g., monthly) returns. Algebraically, it can easily be shown that if the profit factor calculation were applied to monthly returns, the profit factor would equal GPR + 1